

Teacher Guide

MotionLab CubeSat Mission

STEMSat Educational CubeSat Program

1. Mission Overview

In this activity, students investigate how **motion, vibration, and impact forces affect a spacecraft system**. Students build and operate a CubeSat-inspired device equipped with a **triple-axis accelerometer, piezo vibration sensor, and force sensitive resistor (FSR)**. These sensors allow the CubeSat to measure motion, detect vibration, and sense applied forces.

Students conduct experiments such as **drop tests, vibration tests, and applied force tests** to explore how forces influence motion and how spacecraft must be designed to survive dynamic environments. Sensor data is transmitted through the **HC-05 Bluetooth module**, simulating how real satellites send telemetry data to a ground station.

Through testing and data analysis, students explore the relationship between **force, mass, motion, and energy**, while also applying the **engineering design process** to evaluate and improve their CubeSat structure.

2. Target Grade Level

Grades: **6–8 (Middle School)**

3. Time Requirements

Activity	Time
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Mission introduction	10 minutes
Hardware assembly	20 minutes
Code upload and sensor verification	10 minutes
Motion experiments	25 minutes
Data analysis and discussion	15 minutes
Engineering redesign discussion	10 minutes

Total Time: ~90 minutes

This activity can also be divided into **two class periods**.

4. Learning Objectives

By the end of the activity, students will be able to:

1. Explain how motion sensors measure acceleration and vibration
 2. Investigate how applied forces affect motion
 3. Interpret sensor data collected during experiments
 4. Understand why spacecraft must be designed to withstand launch and vibration forces
 5. Apply engineering design principles to improve a system
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5. NGSS Alignment

Engineering Design

Standard	Application
MS-ETS1-2	Students compare results from multiple motion tests and evaluate system performance.
MS-ETS1-3	Students analyze sensor data to identify patterns and design improvements.

MS-ETS1-4	Students build and test a CubeSat model and refine their design through experimentation.
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Science Standards

Standard	Application
MS-PS2-2	Students investigate how forces and mass affect motion using acceleration measurements.
MS-PS3-1	Students interpret data to understand relationships between motion and kinetic energy.

6. Materials

Per Student Group (2–4 students)

- Arduino Uno
 - Triple-axis analog accelerometer
 - Piezo sensor
 - HC-05 Bluetooth module
 - Female to Female Jumper wires
 - Cable holders (optional)
 - USB cable
 - Computer with Arduino IDE
 - CubeSat frame
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Instructor Materials

- Pre-installed Arduino libraries
 - Pre-tested sensors
 - Printed mission guides for each student
 - Foam pads/soft mat for drop tests
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7. Pre-Class Setup

Install Arduino IDE

Install on all computers.

Download:

<https://www.arduino.cc/en/software>

Install Required Libraries

MotionLab uses minimal libraries:

- SoftwareSerial (for Bluetooth communication)

Note:

- This library is often already included with the Arduino IDE
- No additional sensor libraries are required

How to install/check:

1. Open Arduino IDE
2. Go to Tools → Manage Libraries
3. Search: SoftwareSerial
4. Install if not already installed

Test Components / Make Example Model

Before class:

1. (Optional but highly recommended) Pre-attach wiring to sensors (refer to wiring instructions in the mission guide) and optionally use a 3D printed connector holder to hold wires in place and keep students from mixing up wiring for sensors when connecting to the Arduino.
2. Follow the wiring instructions in the Mission Guide to attach sensors and upload code to one Arduino to ensure everything is working/to create an example model for students
3. Verify readings appear in the serial monitor
4. Verify that motion and force readings change when sensors are moved or pressed
5. Pre-check that the Bluetooth module can connect and print out values wirelessly to the serial monitor

Prepare Test Stations

Set up safe areas for experiments such as:

- Drop testing surface (soft mat or foam pad)
 - Table for vibration tests
 - Small weights for mass testing
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8. Running the Activity

Step 1: Introduce the Mission

Ask students:

- Why do rockets and satellites experience large forces during launch?
- Why do engineers test spacecraft for vibration?
- What types of sensors might measure motion?

Explain that students will simulate **spacecraft testing using their CubeSat system**.

Step 2: Assemble the CubeSat

Students connect the accelerometer and piezo sensor to the Arduino.

Instructor checks wiring before power is applied.

Step 3: Upload the Code

Students upload the mission code and open the Serial Monitor.

Confirm that values change when:

- The CubeSat is tilted
 - The piezo sensor is tapped
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9. Motion Experiments

Students perform **three motion experiments**.

Suggested experiments:

Test	Purpose
Drop Test	Simulate impact or landing
Shake Test	Simulate launch vibration
Push Test	Measure applied force

Students record sensor readings during each test.

10. Expected Results

Typical observations:

Sensor	Observation
Accelerometer	Values change when CubeSat tilts or accelerates
Piezo Sensor	Spikes when CubeSat is tapped or shaken

Students should observe that **larger motions or impacts produce larger sensor values**.

11. Data Analysis

Students analyze:

- Which test produced the highest acceleration
- Which produced the strongest vibration signal
- Which produced the highest force reading

Instructor discussion prompts:

- Why did the drop test produce large values?
 - Why do engineers measure vibration in spacecraft?
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12. Engineering Design Discussion

Students evaluate the CubeSat design.

Possible design improvements:

- Add foam padding
- Reinforce frame structure
- Move sensors to different locations
- Secure components more tightly

This reinforces the **engineering design cycle**.

13. Troubleshooting Guide

Problem: Accelerometer values do not change

Possible cause:

- Incorrect wiring

Fix:

- Verify analog pins A0–A2.
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Problem: Piezo sensor does not detect vibration

Possible cause:

- Poor electrical connection

Fix:

- Ensure signal wire is connected to A3.

Problem: Force sensor does not change values

Possible cause:

- Insufficient pressure

Fix:

- Apply stronger pressure or adjust wiring.
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14. Safety Considerations

- Perform drop tests over soft surfaces
 - Avoid dropping electronics on hard floors
 - Keep wires secured during experiments
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15. Extension Activities

Extension 1: Mass Investigation

Students add small weights to the CubeSat and observe changes in motion.

Extension 2: Launch Simulation

Students shake the CubeSat to simulate rocket vibration.

Extension 3: Structural Testing

Students redesign the CubeSat frame to reduce vibration readings.

16. Real-World Space Connection

Spacecraft experience intense forces during launch and deployment. Engineers test satellites using vibration tables and motion sensors to ensure spacecraft can survive these environments.

Similar sensors are used in:

- Rocket launch testing
- Spacecraft vibration testing
- Satellite deployment systems

The MotionLab mission allows students to simulate these real engineering tests.

17. Mission Success Criteria

The mission is successful if:

- CubeSat sensors detect motion, vibration, and force
- Students perform multiple motion experiments
- Students interpret motion data
- Students propose improvements to the CubeSat design